CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. The element having the highest first ionization enthalpy is

(1) C (2) N

(3) Al (4) Si

Answer (2)

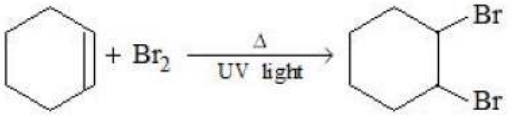
Sol. In general ionisation enthalpy increases on moving left to right in the period and decreases on moving top to bottom in group. Hence 1st ionisation enthalpy is highest for N.

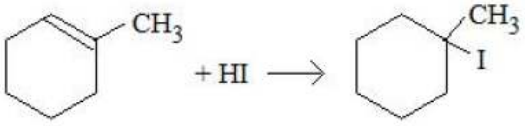
62. Which of the following reaction is correct?

(1) $C_2H_5CONH_2 + Br_2 + NaOH \longrightarrow$

$C_2H_5CH_2NH_2 + Na_2CO_3 + NaBr + H_2O$

(2) $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2,\ 0^oC} CH_3CH_2CH_2OH + N_2 + HCl$

(3) 

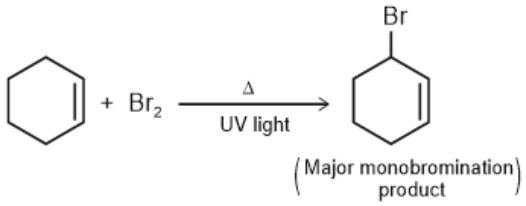
(4) 

Answer (4)

Sol. The correct organic products of given reactions are as follows

1. $C_2H_5CONH_2 + Br_2 + NaOH \rightarrow C_2H_5NH_2$

2. $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2,\ 0^oC} CH_3CH_2CH_2OH$

3. 

4. 

Hence option (4) is correct.

63. A reagent which gives brilliant red precipitate with Nickel ions in basic medium is

(1) Dimethyl glyoxime

(2) Meta-dinitrobenzene

(3) Neutral $FeCl_3$

(4) Sodium nitroprusside

Answer (1)

Sol. Ni^{2+} + dimethyl glyoxime $\xrightarrow{\text{Basic medium}}$ $[Ni(dmg)_2]$ (Cherry red or brilliant red)

64. The correct IUPAC name of K_2MnO_4 is

(1) Potassium tetraoxopermanganate (VI)

(2) Potassium tetraoxidomanganate (VI)

(3) Dipotassium tetraoxidomanganate (VII)

(4) Potassium tetraoxidomanganese (VI)

Answer (2)



Sol. IUPAC name of K_2MnO_4 is

Potassium tetraoxidomanganate (VI)

65. Given below are two statements :

Statement I : Fluorine has most negative electron gain enthalpy in its group.

Statement II : Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the **most appropriate** from the options given below.

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

Answer (4)

Sol. $-\Delta_{eg}H$: S > Se > Te > Po > O $\Rightarrow$ 16th group

$-\Delta_{eg}H$: Cl > F > Br > I $\Rightarrow$ 17th group

66. Alkyl halide is converted into alkyl isocyanide by reaction with

(1) KCN (2) NaCN

(3) NH_4CN (4) AgCN

Answer (4)

Sol. $RX + AgCN \longrightarrow \underset{\text{(Major)}}{RNC} + AgX$

67. Chromatographic technique/s based on the principle of differential adsorption is/are

A. Column chromatography

B. Thin layer chromatography

C. Paper chromatography

Choose the **most appropriate** answer from the options given below:

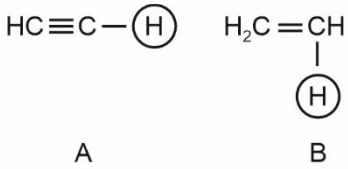
(1) A only (2) C only

(3) B only (4) A & B only

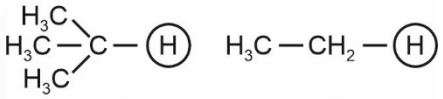
Answer (4)

Sol. Column chromatography and thin layer chromatography is based on principle of differential adsorption while paper chromatography is a type of partition chromatography.

68. The ascending acidity order of the following H atoms is



A B



C D

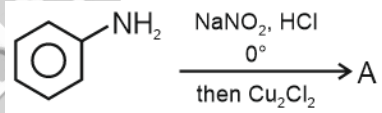
(1) D < C < B < A (2) A < B < C < D

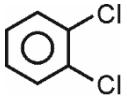
(3) C < D < B < A (4) A < B < D < C

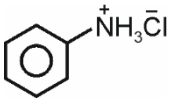
Answer (3)

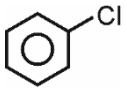
Sol. Order of acidic strength of H attached to carbon : $C_{sp} > C_{sp^2} > C_{sp^3}$. Further differentiation is based on electronic effect of substituents, hence correct order is C < D < B < A.

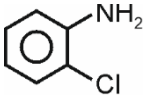
69. The product A formed in the following reaction is



(1) 

(2) 

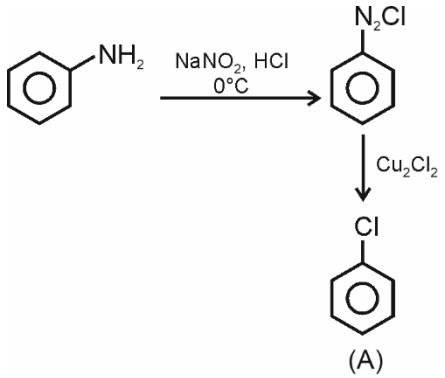
(3) 

(4) 

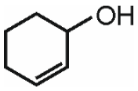
Answer (3)



Sol.



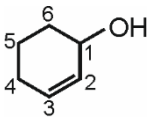
(A)

70. According to IUPAC system, the compound  is named as

(1) 1-Hydroxyhex-2-ene (2) Cyclohex-1-en-2-ol

(3) Cyclohex-1-en-3-ol (4) Cyclohex-2-en-1-ol

Answer (4)

Sol.  is Cyclohex-2-en-1-ol

71. Match List-I with List-II.

	List-I (Bio Polymer)		**List-II (Monomer)**
A.	Starch	I.	Nucleotide
B.	Cellulose	II.	α-glucose
C.	Nucleic acid	III.	β-glucose
D.	Protein	IV.	α-amino acid

Choose the correct answer from the options given below:

(1) A → I, B → III, C → IV, D → II

(2) A → IV, B → II, C → I, D → III

(3) A → II, B → III, C → I, D → IV

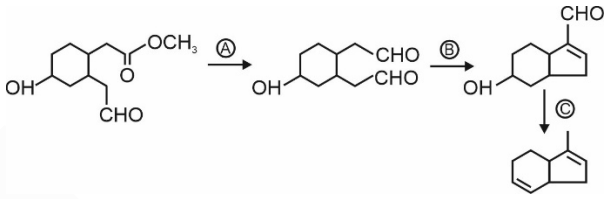
(4) A → II, B → I, C → III, D → IV

Answer (3)

Sol.

Biopolymer	Monomer
Starch	α-D-glucose
Cellulose	β-D-glucose
Nucleic acid	Nucleotide
Protein	α-amino acid (Natural amino acids)

72. Identify the reagents used for the following conversion



(1) A = DIBAL-H, B = $NaOH_{(alc)}$, C = Zn/HCl

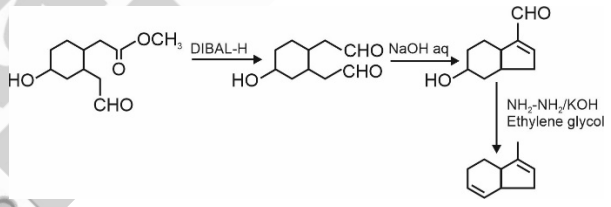
(2) A = DIBAL-H, B = $NaOH_{(aq)}$, C = NH_2-NH_2/KOH, ethylene glycol

(3) A = $LiAlH_4$, B = $NaOH_{(alc)}$, C = Zn/HCl

(4) A = $LiAlH_4$, B = $NaOH_{(aq)}$, C = NH_2-NH_2/KOH, ethylene glycol

Answer (2)

Sol.



73. Phenol treated with chloroform in presence of sodium hydroxide, which further hydrolyzed in presence of an acid results

(1) Benzene-1, 3-diol

(2) Salicylic acid

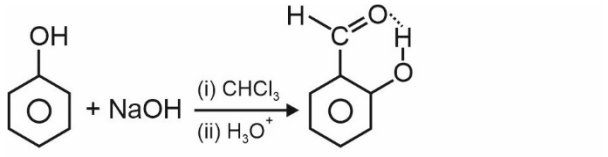
(3) 2-Hydroxybenzaldehyde

(4) Benzene-1, 2-diol

Answer (3)



Sol.



2-Hydroxybenzaldehyde

is Reimer Tiemann reaction.

74. Anomalous behavior of oxygen is due to its

(1) Small size and low electronegativity

(2) Large size and high electronegativity

(3) Large size and low electronegativity

(4) Small size and high electronegativity

Answer (4)

Sol. Anomalous behavior of oxygen is due to small size and high electronegativity as compared to the elements of group 16.

75. Match List-I with List-II.

	List-I (Compound)		**List-II (pK_a value)**
A.	Ethanol	I.	10.0
B.	Phenol	II.	15.9
C.	m-Nitrophenol	III.	7.1
D.	p-Nitrophenol	IV.	8.3

Choose the correct answer from the options given below:

(1) A → III, B → IV, C → I, D → II

(2) A → IV, B → I, C → II, D → III

(3) A → I, B → II, C → III, D → IV

(4) A → II, B → I, C → IV, D → III

Answer (4)

Sol. Order of acidic strength of H attached to Oxygen depend on electronic effect of substituents attached to Oxygen.

pK_a: Ethanol > Phenol > m-Nitrophenol > p-Nitrophenol is in the order of electro withdrawing effect/strength of substituent.

76. Match List I with List II.

List I (Spectral Series for Hydrogen)		**List II (Spectral Region/Higher Energy State)**	
A.	Lyman	I.	Infrared region
B.	Balmer	II.	UV region
C.	Paschen	III.	Infrared region
D.	Pfund	IV.	Visible region

Choose the correct answer from the options given below:

(1) A-I, B-III, C-II, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-IV, C-III, D-I

Answer (4)

Sol. Correct match is as follows:

Spectral series for hydrogen	**Spectral region/Higher Energy state**
Lyman	UV region
Balmer	Visible region
Paschen	Infrared region
Pfund	Infrared region



77. Which of the following acts as a strong reducing agent? (Atomic number: Ce = 58, Eu = 63, Gd = 64, Lu = 71)

(1) Gd^{3+} (2) Eu^{2+}

(3) Lu^{3+} (4) Ce^{4+}

Answer (2)

Sol. Eu^{2+} is a strong reducing agent changing to the common +3 state while all other are provided at their higher oxidation states.

78. Which of the following statement are correct about Zn, Cd and Hg?

A. They exhibit high enthalpy of atomization as the d-subshell is full.

B. Zn and Cd do not show variable oxidation state while Hg shows +I and +II.

C. Compounds of Zn, Cd and Hg are paramagnetic in nature.

D. Zn, Cd and Hg are called soft metals.

Choose the **most appropriate** from the options given below:

(1) B, D only

(2) C, D only

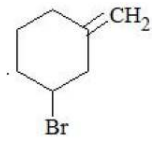
(3) B, C only

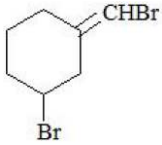
(4) A, D only

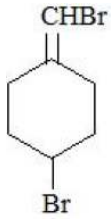
Answer (1)

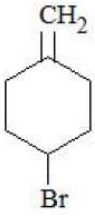
Sol. Zn, Cd and Hg are not regarded as transition elements as they have $(n-1)d^{10}ns^2$ configuration so they have low enthalpy of atomization and their compounds are diamagnetic in general.

79. Which one of the following will show geometrical isomerism?

(1) 

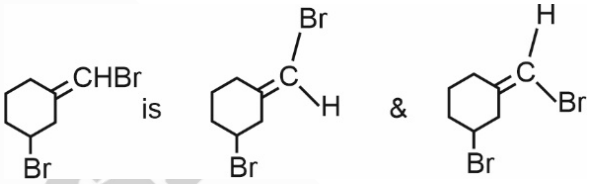
(2) 

(3) 

(4) 

Answer (2)

Sol. The geometrical isomers of



While in case of other structures geometrical isomerism is not possible.

80. On passing a gas, 'X', through Nessler's regent, a brown precipitate is obtained. The gas 'X' is

(1) Cl_2

(2) NH_3

(3) H_2S

(4) CO_2

Answer (2)

Sol. Nessler's reagent is used for the test of NH_3.

$$NH_3 + HgI_4^{2-} \xrightarrow{OH^-} \underset{\text{brown ppt}}{HgO \cdot Hg(NH_2)I\downarrow}$$



SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

81. Standard enthalpy of vapourisation for CCl_4 is 30.5 kJ mol^{-1}. Heat required for vapourisation of 284 g of CCl_4 at constant temperature is ____kJ.

(Given molar mass in g mol^{-1}; C = 12, Cl = 35.5)

Answer (56)

Sol. Moles of $CCl_4 = \frac{284}{154} = 1.8$

Heat required = 1.8 mol × 30.5 kJ mol^{-1}

= 56.24 kJ

$\simeq$ 56 kJ

82. The oxidation number of iron in the compound formed during brown ring test for NO_3^- ion is _____.

Answer (1)

Sol. Brown ring complex is $[Fe(H_2O)_5NO]^{2+}$. It has Fe at +1 oxidation state that can be explained by magnetic moment data of complex. It is an example of ligand to metal charge transfer that is why Fe has +1 and NO has +1 oxidation state.

83. The following concentrations were observed at 500K for the formation of NH_3 from N_2 and H_2. At equilibrium ; $[N_2] = 2 \times 10^{-2}$ M, $[H_2] = 3 \times 10^{-2}$ M and $[NH_3] = 1.5 \times 10^{-2}$ M. Equilibrium constant for the reaction is_____.

Answer (417)

Sol. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{[1.5\times10^{-2}]^2}{[2\times10^{-2}][3\times10^{-2}]^3}$$

= 416.67 $\simeq$ 417

84. A constant current was passed through a solution of $AuCl_4^-$ ion between gold electrodes. After a period of 10.0 minutes, the increase in mass of cathode was 1.314g. The total charge passed through the solution is ____ × 10^{-2} F.

(Given atomic mass of Au = 197)

Answer (2)

Sol. Moles of Au deposited = $\frac{1.314}{197}$

Equivalents of Au deposited = Equivalents of charge conducted

$\frac{1.314\times3}{197}$ = Charge in F

= 0.02001

$\simeq 2 \times 10^{-2}$

As one equivalent charge is one faraday.

85. The half-life of radioisotope bromine-82 is 36 hours. The fraction which remains after one day is _____ × 10^{-2}.

(Given antilog 0.2006 = 1.587)

Answer (63)



Sol. $\lambda = \frac{2.303}{t}\log\frac{N_0}{N_t}$

$$\frac{2.303\times.301}{36} = \frac{2.303}{24}\log\frac{N_0}{N_t}$$

$$\log\frac{N_0}{N_t} = 0.2006$$

$$\frac{N_0}{N_t} = 1.587$$

$$\frac{N_t}{N_0} = 0.6301$$

$= 63 \times 10^{-2}$

86. The total number of anti bonding molecular orbitals, formed from 2*s* and 2*p* atomic orbitals in a diatomic molecules is

Answer (4)

Sol. In the formation of diatomic molecule, the total number of 2*s* and 2*p* orbitals participate is 8.

So, 8 molecular orbitals will form four of which will be antibonding, hence answer will be 4.

87. Molality of 0.8 M H_2SO_4 solution (density 1.06 cm^{-3}) is _____ $\times 10^{-3}$ m.

Answer (815)

Sol. $$\text{Molality} = \frac{\text{Molarity} \times 10^3}{(1000 \times d) - (\text{Molarity} \times \text{Molar mass})}$$

$$= \frac{0.8\times10^3}{1000\times1.06 - 0.8\times98}$$

$= 0.81499$

$= 815 \times 10^{-3}$

88. The total number of molecules with zero dipole moment among CH_4, BF_3, H_2O, HF, NH_3, CO_2 and SO_2 is _____.

Answer (3)

Sol. CH_4, BF_3, CO_2 are nonpolar while H_2O, HF, NH_3 and SO_2 are polar that can be explained by their geometry.

CH_4 : Tetrahedral, BF_3 : Trigonal planar,

CO_2 : Linear

89. If 50 mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is _____g.

Answer (4)

Sol. Let molarity of NaOH is M.

Equivalents of $H_2C_2O_4$ = Equivalents of NaOH

$0.5 \times 50 \times 10^{-3} \times 2 = M \times 25 \times 10^{-3} \times 1$

$M = 2$

Mass of NaOH in 50 mL of such solution

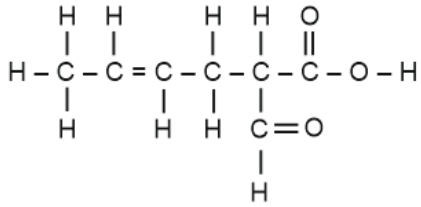
$= 2 \times 50 \times 10^{-3} \times 40$

$= 4$ g

90. The total number of 'Sigma' and 'Pi' bonds in 2-formylhex-4-enoic acid is _____.

Answer (22)

Sol. 2-formylhex-4-enoic acid is



Sigma bonds : 19

pi bonds : 3

❐ ❐ ❐

